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Technical and timing characteristics of badminton men's single: comparison between groups and play-offs stages in 2016 Rio Olympic Games

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ABSTRACT

Badminton match is characterised by intermittent efforts of high intensity and short-duration rallies interspersed by periods of rest. The analyses of technical and timing variables provide important information for understanding the match. The aim of this study was to characterise the technical and timing variables of badminton men's singles matches and to compare the characteristics between groups and play-offs stages in 2016 Olympic Games. All men's single matches (groups = 43 and play-offs = 13) from Rio 2016 were used to carry out the post-event analysis. The comparison between phases showed that total time, pause time, points played and shots/rally were significantly higher in the play-offs phase. The frequency of serve, net, smash and total shots was also significantly higher in the play-offs. Only drive presented higher frequency in the group phase. We can conclude that play-offs were more intense than the group stage. Research in this area is relevant for understanding the match and providing information for training planning. The results of this study are important to set the appropriate targeting of workloads in view of the requirements of a match.

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Introduction

Badminton is a sport that uses the racket to hit a shuttle, played between two (or four) players with the goal of driving the shuttle to a place where the opponent would not be able to successfully hit it back (Lees, 2003). Originally from China and developed in England (Tan, Ting, & Lau, 2016), nowadays there are more than 220 million regular practitioners of Badminton, being one of the most practiced sports in the world (Li, Zhang, Wan, Wilde, & Shan, 2017). Badminton gained prominence after its insertion in the Olympic Games of Barcelona 1992 and, since then, the sport is disputed in five categories: men's single, women's single, men's doubles, women's doubles and mixed doubles.

The badminton match is characterised by high intensity and short-duration intermittent efforts, interspersed by periods of rest (Abián-Vicén, Del Coso, González-Millán, Salinero, & Abián, 2012; Manrique & Gonzalez-Badillo, 2003). During these efforts, also known as

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rallies, numerous motor actions occur through displacements by the court. The number of strokes of a match can vary a lot due to the several possibilities of tactical actions (Seth, 2016). Thus, technical and timing variables are frequently investigated in performance analysis studies of badminton matches (Phomsoupha & Laffaye, 2015).

Timing variables comprises playing time, rally time, pause time and effective playing time. The technical analysis corresponds to the quantification of the motor actions performed by the athletes (Drust, 2010; Lees, 2003). The knowledge of these characteristics could help to understand the metabolic and motor requirements of the match (Phomsoupha & Laffaye, 2015).

Once the structure of the match is understood, it is possible to plan the training according to the requirements of the game in order to improve the athletes' performance (Abdullahi & Coetzee, 2017; Abián, Castanedo, Feng, Sampedro, & Abian-Vicen, 2014; Li et al., 2017; O'Donoghue, 2009; Ting, Tan, & Lau, 2016). Research that explores the dynamics of the match allows monitoring the match evolution and makes it possible to keep updated the references for training. Laffaye, Phomsoupha, and Dor (2015) identified differences in the intensity of men's single matches during the Olympic finals from 1992 (Barcelona) to 2012 (London) showing an increased shots frequency (+34%), decreased effective playing time (−34.5%) and working density (−58.2%). However, the number of research involving badminton match analysis is inversely proportional to its popularity (Li et al., 2017). Since it has become an Olympic sport (Barcelona 1992), few studies analysed the badminton match in the Olympic context (Laffaye et al., 2015).

Studies that analysed technical and timing variables in badminton matches at the Olympic Games considered only isolated matches. Furthermore, the studies showed different aims, such as: to record the evolution in Olympic finals in the men's category (Laffaye et al., 2015), to compare men's matches between two Olympic Games (Abián et al., 2014) and to compare matches between men's and women's category (Abian-Vicen, Castanedo, Abian, & Sampedro, 2013; Valldecabres, Benito, Casal, & Pablos, 2017). To the best of our knowledge, there is no study that analysed the technical and timing variables in world elite badminton considering all the matches in a same edition of Olympic Games. Likewise, we could find no studied that compared technical and timing characteristics between the two phases (groups and play-offs) in the same event.

According to the classification criteria for the Olympic Games, this event assembles the most successful athletes of the sport at that time. Therefore, the aim of this study was to characterise the technical and timing variables of the badminton game in the men's category of the 2016 Olympic Games and compared the characteristics between the two phases of the competition (group and play-offs).

Methods

All men's single matches (n = 56) from the 2016 Olympic Games were analysed. The competition was held in two moments. The group stage (43 matches) played in a Round-Robin system, acted as qualification for the play-off phase (13 matches). All matches were played according to official badminton rules.

Procedures

Official videos recorded by the organisation of the 2016 Olympic Games were used to carry out the analysis of the matches using Badminton Observational Tool validated by

Valldecabres et al. (2017). Reproduction was carried out using media player software and all variables were recorded in Microsoft Excel spreadsheets. Two qualified sport scientists, with experience in badminton, were trained to perform all analysis adopting the same criteria.

Two changes took place in the last edition of the Olympic Games, one of which was the reduction of the number of participants by the Olympic Committee (NOC) and the other change was the insertion of the instant review system for line calls which may have influenced the characteristics of the game, since each player had the right to request two revisions per game.

Variables

The collected variables were divided into technical and timing. The timing variables were: total time (the length of the match beginning with the first point and going through the last point of the match), total rally duration (the sum of all rally time of the match), rally time (time elapsed from the serve until the end of the point), total rest time (the sum of all rest time of the match), rest time (time elapsed from the point ending until the beginning of the following point) (Phomsoupha & Laffaye, 2015). Work density (ratio of performance time to rest time), % played, shots/rally and shots/second were non-dimensional, calculated from the timing and technical variables.

The technical variables, and the respective adopted criteria, are presented below:

Serve – the moment the player touches the shuttle at the beginning of the point

Lob – shot towards the back of opponent's court with a raining trajectory (Phomsoupha & Laffaye, 2015)

Drop – it is a downward trajectory smooth overhead shot to the front of the court (Phomsoupha & Laffaye, 2015)

Net – precise shot near the net, which includes push, kill and brush (Abian-Vicen et al., 2013)

Smash – aggressive overhead shot with downward trajectory (Phomsoupha & Laffaye, 2015)

Clear – overhead shot to the back of court in a rising trajectory of the shuttle (Abian-Vicen et al., 2013)

Drive – a powerful shot made at middle body height from the middle of the court and has a flat trajectory (Phomsoupha & Laffaye, 2015)

Five matches (randomly selected) were analysed by a third evaluator to verify the reproducibility of data collection. The Kappa (K) values ranged from .85 to .95 for the technical and from .86 to .93 for the temporal variables.

Statistical analysis

The descriptive statistics (mean, minimum and maximum standard deviation) were used to present the data collected. The Shapiro–Wilk statistical test was used to analyse the normality of the data. Most of the variables did not follow a normal distribution and, in these cases, Mann–Whitney test was used for the comparison between play-offs and group stages. When the variable showed a normal distribution in both phases, t-test for independent samples was used for the comparison. The Cohen's d effect size (ES) was adopted to analyse the magnitude of the effect. The Rhea (2004) classification for highly trained individuals was used to analyse the ES. It was considered: trivial <.25; .25 < small <.50; .50 < moderate <1.0;

large >1.0. The Friedman test was used to compare the frequency between different types of shots. When differences were found, Dunn's *post hoc* was used to identify them. The Spearman coefficient was used to analyse the correlation between variables. All analyses were performed in the MATLAB 2010 programme. The level of significance was set at .05.

Results

Match characteristics

Figure 1 shows the average shots frequencies per match. The net and lob were the most frequent shots and drive was the least frequent compared to the other strokes ($p < .05$). Serve, drop, smash and clear did not present significant differences amongst them.

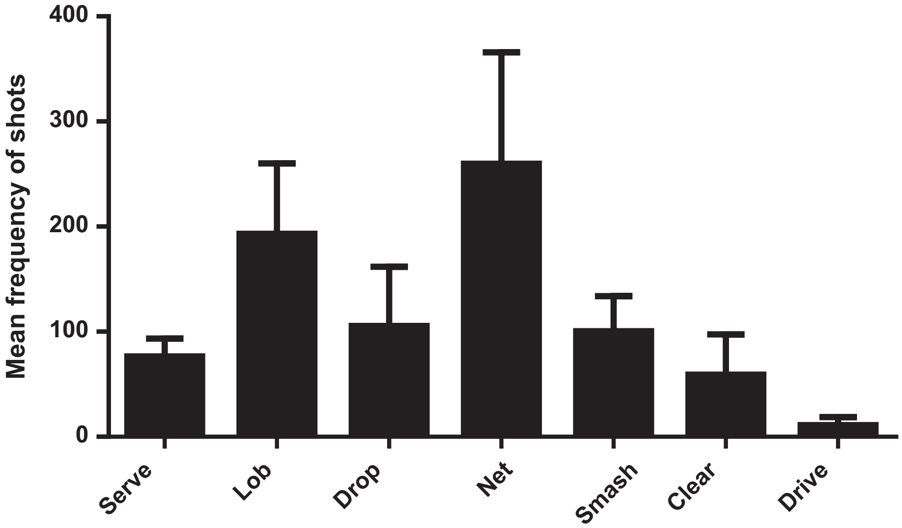

Figure 1. Average frequencies of shots per match in the men's simple badminton matches in 2016 Olympic Games.

Figure 2 shows the distribution of rest and playing times of all rallies. The games occur most frequently (73%) in rallies of up to 12 s and, on the other hand, pauses occur more frequently from that time interval. The mean playing and rest time were 9.5 (±7.9s) and 24.9 (±16.1s), respectively.

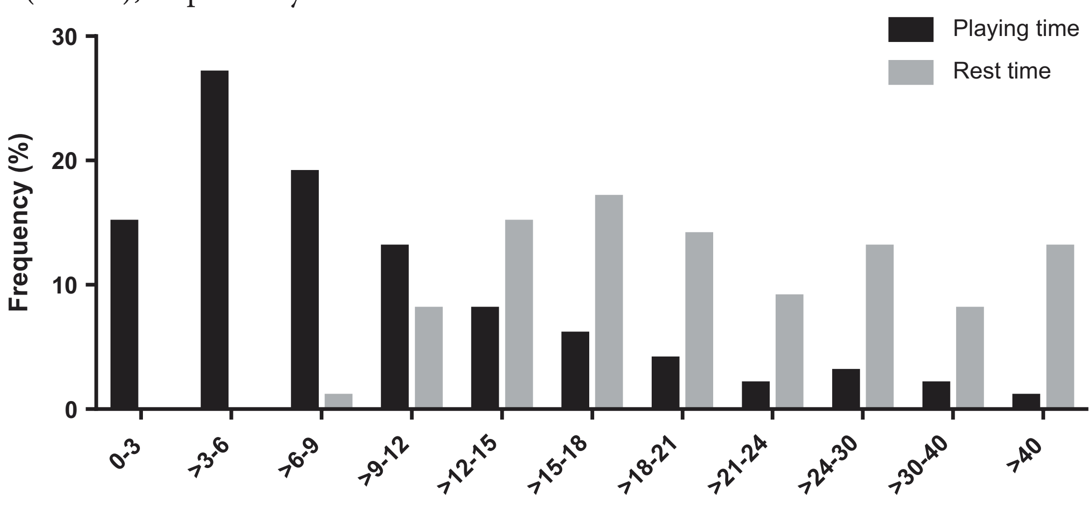

Figure 2. Frequency of playing intervals and recovery in all rallies in 2016 Olympic Games.

Figure 3 shows the frequency distribution of shots per rally. It is noted that in 80% of rallies, the frequency of shots is lower than 15.

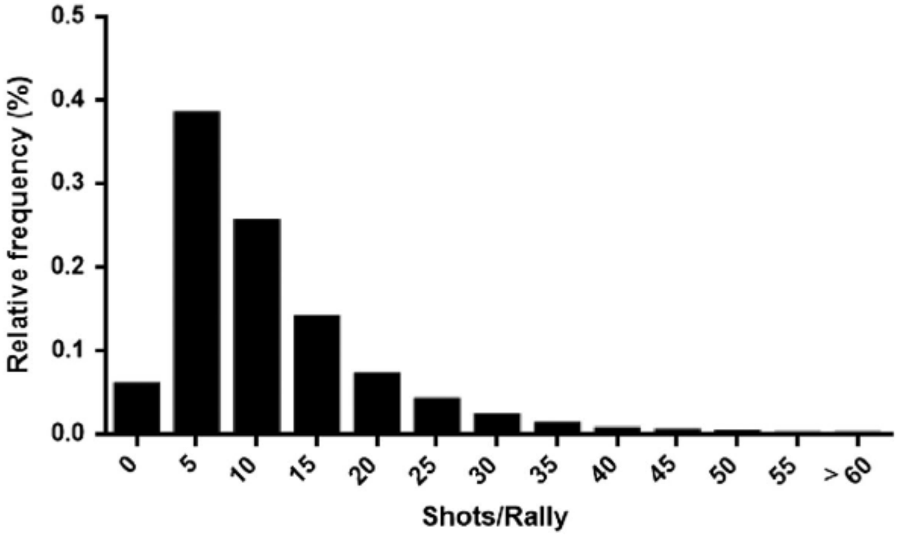

Figure 3. Distribution of shots per rally frequency in the 2016 Olympic Games men's badminton games.

Comparison between the phases

Table 1 shows the characterisation of the matches and the comparisons of the variables between the group and play-offs stages. Total time, rest time, points played and shots/rally were significantly higher in the play-offs when compared to the group stage. Regarding the technical variables, serve, net, smash and total shots were significantly higher in the eliminatory phase. Only drive was higher in group stage.

Table 1. Timing and technical characteristics of male badminton matches in group and play-offs in 2016 Olympic Games.

		All Matches (n = 56)	Groups (n = 43)	Play-off (n = 13)	p-value	ES
Timing variables	Total time (s)	2745.5 (928.9)	2522.0 (721.0)	3464.0 (1136.0)	<.01*	.99
	Total rally duration (s)	730.5 (245.5)	677.7 (188.3)	907.7 (328.3)	.05	.86
	Total rest time (s)	2014.9 (706.7)	1845.0 (560.4)	2557 (829.2)	<.01*	1.00
	% Played	26.6 (.1)	27.1 (3.1)	26.0 (3.1)	.21	.33
	Work density	.37 (.1)	.37 (.1)	.35 (.1)	.20	.2
	Points played	76.6 (16.9)	74.1 (16.2)	84.9 (16.9)	.02*	.66
	Shots/rally	10.4 (1.9)	10.1 (1.7)	11.5 (2.2)	.04*	.71
	Shots/seconds	1.1 (.0)	1.1 (.0)	1.1 (.0)	.89	.71
Shots	Serve	76.6 (16.9)	74.0 (16.2)	84.9 (16.9)	.02*	.66
	Lob	193.1 (66.9)	179.5 (50.3)	238.2 (93.6)	.14	.79
	Drop	105.6 (56.3)	95.4 (35.3)	139.1 (92.6)	.19	.63
	Net	259.9 (105.7)	229.4 (79.4)	361.2 (121.3)	<.01*	1.29
	Smash	100.6 (33.2)	96.2 (32.7)	115.3 (31.8)	.03*	.59
	Clear	59.0 (38.2)	60.7 (40.5)	54.2 (30.3)	.84	.17
	Drive	11.3 (7.6)	12.3 (7.8)	7.8 (6.2)	.03*	.63
	Total shots	806.4 (265.3)	747.6 (205.1)	1001.0 (349.3)	.04*	.88

Notes: Data are presented as mean (std); ES – effect size.
*$p < .05$.

Differences were found between the two phases of the competition in: total time, total rally duration and total rest duration, considering the effect size. In relation to the shots, only total shots and nets presented difference between the phases.

The mean of all the rallies time in the group phase, 9.3 (±14.3s), was significantly lower than the play-offs, 10.7 (±8.7s) ($p < .01$, ES = .12). Similarly, the mean rest time in group phase, 25.6 (±21.4s), was significantly lower than the play-offs, 30.5 (±23.2s) ($p < .01$, ES = .23).

Discussion

The aim of this study was to characterise the technical and timing variables of the badminton game in the men's category of the 2016 Olympic Games and to compare the characteristics between the two phases of the competition (group and play-offs). The results showed that net and lob were the most frequent shots and that most of the rallies take less than 12s. The comparison between phases showed longer total time, total rest time, points played and shots/rally in the play-offs stage compared to the group stage. Regarding the technical variables, serve, net, smash and total shots were significantly higher in the eliminatory phase and only drive was higher in the group's stage.

Timing analysis

Regarding the timing variables, we found longer duration (2745.5 ± 928.9s) in male single matches in 2016 Olympic Games compared to previous Olympic Games editions (2378.0 ± 387.9 s) (Abian-Vicen et al., 2013) and other international matches (Laffaye et al., 2015). The present study found similar playing times to those of Beijing (2008) and London (2012) Olympic Games (Abián et al., 2014). The longer duration of the matches and the similar playing time compared to previous editions of the games maybe due to the insertion of instant review system used to review line calls in Rio 2016. Regarding the number of played points, the last Olympic Games (Rio 2016) recorded more points (76 points) compared to Beijing 2008 (68 points) (Abian-Vicen et al., 2013). It is important to point out that a greater number of points and the same effective time demands better performance of the athlete to support all efforts from the game.

The rallies and rest duration (9.5 ± 7.9s and 24.9 ± 16.1s, respectively) found in this study is similar to of other Olympic events (Abian-Vicen et al., 2013; Abián et al., 2014) and higher compared to international matches (i.e. 5.5 ± 4.0s in rallies and 11.4 ± 6.0s in rest) (Faude et al., 2007). The long duration the rally assumes a great physical requirement of the athletes. On the other hand, longer pause favours a great recovery between the points. Based on the similar pause time found in this study and in the literature of other Olympic events, we assume that the insertion of the instant review system did not influence the dynamics of the matches.

It is important to analyse the distribution of the rest time and rally time throughout the game. The rally starts with a serve and ends when some athlete makes a mistake. Therefore, the rally time can vary greatly throughout the match. The present study corroborates the literature showing a high frequency of rallies of up to 12 s (Abian-Vicen et al., 2013; Abián et al., 2014; Laffaye et al., 2015). Regarding the rest time, it starts with the player's error and goes until the beginning of the next point, being also a variable that can vary throughout the match. Most of the rest time found in this study lasted more than 12 s, which is also found in the literature (Abian-Vicen et al., 2013; Laffaye et al., 2015; Manrique & Gonzalez-Badillo, 2003).

The results of shots per rally found in this study is similar to those in Beijing and London Olympic Games (Abián et al., 2014). It shows a balance in the dynamics of the men's game at this competition level. However, we found higher values of shots per rally in 2016 Olympic Games compared to simulated games (Chen & Chen, 2011), to matches with the old scoring system (Manrique & Gonzalez-Badillo, 2003) and to matches with young athletes (Ming, Keong, & Ghosh, 2008). This difference can be due to the technical level of the players, since athletes with higher level of expertise have a greater capacity to accelerate the trajectory of the shuttle (Laffaye et al., 2015). Although the result of shots/rally is important, it should not be based on only mean value. In 80% of the cases, rallies are executed up to 15 shots, but there are rallies with smaller frequency of shots per rally than the average of the match.

The variable of work density shows the relationship between effort and pause in the game. The average values found in this study (.37) is similar to the Beijing 2008 (.37) and London 2012 (.39) Olympic Games (Abián et al., 2014). However, the results found in this study are lower compared to the others at international levels, which registered a mean of .4 (Abdullahi & Coetzee, 2017), .49 (Manrique & Gonzalez-Badillo, 2003) and .53 for top national athletes (Cabello, Padial, Lees, & Rivas, 2004). These results suggest that athletes who compete at higher levels cadence more the game, that is, they take more advantage of the recovery time between points.

The comparison between the phases showed that group and play-off stages have different timing characteristics regarding the amount of total time, points played, rally time, total rest time, rest time and the number of strokes/rally. Due to the possibility of elimination during the play-off stage, it is assumed a greater contest in this stage. Therefore, more points are played, more strokes are performed and more rally time is required, resulting in the larger volume of the match. This higher demand, supposedly, requires also greater total rest time. Experienced athletes rest more between rallies for slow down momentum of the match. This moment is then optimised for a more effective recovery, with athletes ready for the next point (Abdullahi & Coetzee, 2017).

Technical analysis

Previous studies have already analysed some technical aspects of the badminton match (Abdullahi & Coetzee, 2017; Abian-Vicen et al., 2013; Abián et al., 2014; Ming et al., 2008; Tong & Hong, 2000). However, each study focused on specific aspects of the match and analysed different levels of competitions. For that reason, it is difficult to discuss our findings. The average number of hits per match found in our study was 806.4, with the highest frequency of the net, followed by lob, drop, smash, clear and drive. The values show that men's single has a great participation of shots near the net, where top badminton players produces more effective shots (Tong & Hong, 2000).

Ming et al. (2008) also found that lob and net were the most executed shots in matches of young badminton athletes. On the other hand, Abdullahi and Coetzee (2017) showed an average of 444.3 ± 101.1 strokes per match with the highest frequency of drives, followed by clear, serves, smash and net in the African Badminton Championship matches. Tong and Hong (2000) found that the most executed shots in a badminton match were lob, smash, net and clear. It is noted that this distribution of shots throughout a match of badminton is still a controversy in the badminton literature.

It is interesting to note that smash, considered the most efficient offensive shot in badminton, represents only 12.5% of the total shots performed in a match, contrary to studies that highlight this blow as being predominant in badminton matches (Jaitner & Gawin, 2010; Tong & Hong, 2000). Tong and Hong (2000) highlights the fact that the smash is a very important shot in badminton match, but, it is impossible to smash all the time, as the opponent may not provide such an opportunity. Although, the smash execution demand excellent coordination of all body segments which elicit a high energy demand (Li et al., 2017), so if practiced too often, can raise levels of fatigue (Abdullahi & Coetzee, 2017), which would not be a good strategy for an Olympic event.

In the comparison between the phases of the competition (groups and play-offs), we observed a higher number of points played in the play-offs phase, followed by a significant increase in the number of serves at this stage. In addition, during the play-offs, the number of net, smash and total points are also higher. We highlight the fact that the number of smash increased in the play-off phase, which according to Li et al. (2017) is considered the most effective offensive shot since it can score a point directly or even make the opponent defend passively. Another important point worth mentioning was the significant reduction in the number of drives in the play-offs which, combined with a significant increase in the nets, indicating a preference of the athletes to play near the net in the final stages of the event.

It is important to recognise some limitations of the present study. First, the effectiveness of the shots was not analysed, making it impossible to identify which are most efficient to make a point in a match. Second, neither the distribution of the shots according to the areas of the court was investigated in order to identify which area of play is most exploited in matches of this level. Third, the shuttle's trajectory during the match was not analysed, information that would be useful to understand the changes of direction during the match and the tactical development of the athletes. Fourth, future studies should include the technical analysis of all badminton shots during a match.

In summary, this study carried out the characterisation of the badminton matches of the men's category of the 2016 Olympic Games, regarding the timing and motor actions of a match. The technical and timing characteristics of the two phases of the event were compared showing more intense matches during the play-off stage. Research in this area is relevant for understanding of the match and the training planning, in order to set the appropriate targeting of workloads in view of the requirements of a match.

This type of research feeds coaches and athletes to minimise mistakes in training planning, as they would avoid loading below or beyond the match requirement, which could lead to injury (Gabbett, 2016). Kraemer et al. (2003) report that organised and well-planned training programmes were more efficient in improving sports performance than unplanned programmes, reinforcing the importance of understanding the match.

Disclosure statement

No potential conflict of interest was reported by the authors.

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